

Constants

1.0

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Chapter 1

Constants Library

1.1 Introduction

This library provides a collection of commonly used mathematical constants, unit-conversion macros, SI and IEC prefixes, and data-type-related constants.

The library is organized into three main categories:

- Mathematical constants and precomputed values.
- Unit conversion and measurement prefixes.
- Data type limits, ranges, and floating-point precision values.

1.2 Features

The constants library provides:

- Mathematical constants such as pi and tau.
- Precomputed powers of Euler's number.
- Precomputed square-root and cube-root values.
- Integer type minimum and maximum values.
- Floating-point range and magnitude constants.
- Floating-point machine epsilon values.
- Macros for unit conversions, such as:
 - Length, mass, and time
 - Frequency and period
 - Temperature
 - Angle
 - Rotational speed
- SI decimal prefixes.
- IEC binary prefixes.

1.3 Header Files

The library is provided through the following headers:

- [constants.h](#) - Main header that includes all constant definitions.
- [constants_math.h](#) - Mathematical constants and precomputed values.
- [constants_units.h](#) - Unit conversion macros and SI/IEC prefixes.
- [constants_data_type.h](#) - Data type range and precision constants.

1.4 Mathematical Constants

The mathematical constants header provides commonly used values of pi, tau, Euler's number, square roots, and cube roots.

Examples include:

```
double pi = MATH_PI;
double tau = MATH_TAU;
double e = MATH_e;

double sqrt_2 = MATH_SQRT_2;
double cbrt_2 = MATH_CUBRT_2;
```

Precomputed multiples and sub-multiples of pi are also available, together with powers of Euler's number and roots from 2 through 25.

1.5 Unit Conversion

The unit conversion header provides macros for commonly used engineering and scientific unit conversions.

1.5.1 Length

Examples:

```
double meters = IN_2_M(10.0);
double feet = M_2_FT(2.0);
double miles = M_2_MILE(1609.344);
```

1.5.2 Mass

Examples:

```
double kg = LB_2_KG(10.0);
double lb = KG_2_LB(5.0);
```

1.5.3 Time

Examples:

```
double ms = SEC_2_MS(2.0);
double s = MS_2_SEC(500.0);
```

1.5.4 Frequency and Period

Examples:

```
double period = HZ_2_SEC(100.0);
double freq   = SEC_2_HZ(0.01);
```

1.5.5 Temperature

Examples:

```
double fahrenheit = C_2_F(25.0);
double kelvin     = C_2_K(25.0);
double celsius    = F_2_C(77.0);
```

1.5.6 Angle

Examples:

```
double radians = DEG_2_RAD(90.0);
double degrees = RAD_2_DEG(MATH_PI);
```

1.5.7 Rotational Speed

Examples:

```
double radps = RPM_2_RADPS(1000.0);
double rpm   = RADPS_2_RPM(10.0);
```

1.6 Prefixes

The library provides SI decimal prefixes ranging from QUETTA through QUECTO.

Examples:

```
double mega = MEGA;
double micro = MICRO;
double nano = NANO;
```

IEC binary prefixes are also provided:

```
auto kibi = KIBI;
auto mebi = MEBI;
auto gibi = GIBI;
```

1.7 Data Type Constants

The data type constants header provides minimum and maximum values for integer types together with floating-point range and precision constants.

Examples:

```
auto int_min = INT_32_MIN;
auto int_max = INT_32_MAX;

double epsilon = DOUBLE_EPSILON;
float  small   = FLOAT_SMALL;
```

Floating-point constants include:

- `FLOAT_MIN`
- `FLOAT_MAX`
- `FLOAT_MIN_POS`
- `FLOAT_SMALL`
- `FLOAT_BIG`
- `FLOAT_EPSILON`
- `DOUBLE_MIN`
- `DOUBLE_MAX`
- `DOUBLE_MIN_POS`
- `DOUBLE_SMALL`
- `DOUBLE_BIG`
- `DOUBLE_EPSILON`

1.8 Basic Usage

The complete constants library can be included through the main header:

```
#include "constants.h"

double angle = DEG_2_RAD(90.0);
double value = MATH_PI * 2.0;
double time  = MS_2_SEC(100.0);
```

1.9 Library Organization

The library is divided into the following functional groups:

- Mathematical constants: `constants_math.h`
- Unit conversions and prefixes: `constants_units.h`
- Data type constants: `constants_data_type.h`
- Unified library interface: `constants.h`

1.10 Copyright

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Chapter 2

Constants

Constants

Chapter 3

Directory Hierarchy

3.1 Directories

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Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

include/ constants.h	
Aggregates mathematical, unit, and data type constants	13
include/ constants_data_type.h	
Minimum, maximum, range, and precision constants for data types	14
include/ constants_math.h	
Mathematical constants and precomputed root values	20
include/ constants_units.h	
Unit conversion macros and SI/binary prefixes	34

Chapter 5

Directory Documentation

5.1 doc Directory Reference

5.2 include Directory Reference

Files

- file [constants.h](#)
Aggregates mathematical, unit, and data type constants.
- file [constants_data_type.h](#)
Minimum, maximum, range, and precision constants for data types.
- file [constants_math.h](#)
Mathematical constants and precomputed root values.
- file [constants_units.h](#)
Unit conversion macros and SI/binary prefixes.

Chapter 6

File Documentation

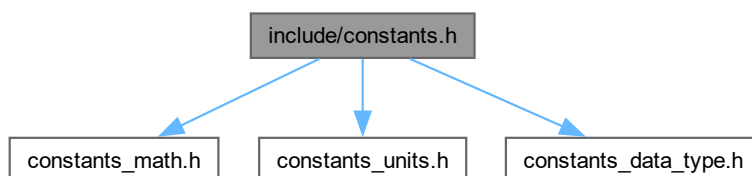
6.1 doc/mainpage.dox File Reference

6.2 include/constants.h File Reference

Aggregates mathematical, unit, and data type constants.

```
#include "constants_math.h"
#include "constants_units.h"
#include "constants_data_type.h"
```

Include dependency graph for constants.h:



6.2.1 Detailed Description

Aggregates mathematical, unit, and data type constants.

Includes the headers that define the library's constant sets.

Definition in file [constants.h](#).

6.3 constants.h

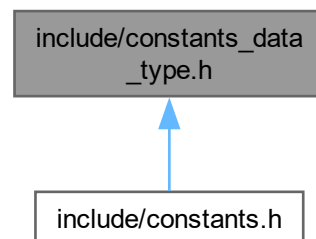
[Go to the documentation of this file.](#)

```
00001
00007
00008 #ifndef CONSTANTS_H
00009 #define CONSTANTS_H
00010
00011 #include "constants_math.h"
00012 #include "constants_units.h"
00013 #include "constants_data_type.h"
00014
00015 #endif
```

6.4 include/constants_data_type.h File Reference

Minimum, maximum, range, and precision constants for data types.

This graph shows which files directly or indirectly include this file:



Macros

- #define [DOUBLE_BIG](#) 1e+300
- #define [DOUBLE_EPSILON](#) 2.220446049250313e-16
- #define [DOUBLE_MAX](#) 1.7976931348623158e+308
- #define [DOUBLE_MIN](#) -1.7976931348623158e+308
- #define [DOUBLE_MIN_POS](#) 2.2250738585072014e-308
- #define [DOUBLE_SMALL](#) 1e-300
- #define [FLOAT_BIG](#) 1e+30f
- #define [FLOAT_EPSILON](#) 1.1920929e-7f
- Floating-point machine epsilon constants.*
- #define [FLOAT_MAX](#) 3.402823466e+38f
- #define [FLOAT_MIN](#) -3.402823466e+38f
- Floating-point range and magnitude constants.*
- #define [FLOAT_MIN_POS](#) 1.175494351e-38f
- #define [FLOAT_SMALL](#) 1e-30f
- #define [INT_16_MAX](#) 32767i16
- #define [INT_16_MIN](#) (-32767i16 - 1)
- #define [INT_32_MAX](#) 2147483647i32

- `#define INT_32_MIN` (-2147483647i32 - 1)
- `#define INT_64_MAX` 9223372036854775807i64
- `#define INT_64_MIN` (-9223372036854775807i64 - 1)
- `#define INT_8_MAX` 127i8
- `#define INT_8_MIN` (-127i8 - 1)

Integer type limits.

- `#define LONG_MAX` 2147483647L
- `#define LONG_MIN` (-2147483647L - 1)
- `#define UINT_16_MAX` 0xffffui16
- `#define UINT_16_MIN` 0x0ui16
- `#define UINT_32_MAX` 0xffffffffui32
- `#define UINT_32_MIN` 0x0ui32
- `#define UINT_64_MAX` 0xffffffffffffffffui64
- `#define UINT_64_MIN` 0x0ui64
- `#define UINT_8_MAX` 0xffui8
- `#define UINT_8_MIN` 0x0ui8
- `#define ULONG_MAX` 0xffffffffUL
- `#define ULONG_MIN` 0x0UL

6.4.1 Detailed Description

Minimum, maximum, range, and precision constants for data types.

Definition in file [constants_data_type.h](#).

6.4.2 Macro Definition Documentation

6.4.2.1 DOUBLE_BIG

```
#define DOUBLE_BIG 1e+300
```

Definition at line 56 of file [constants_data_type.h](#).

6.4.2.2 DOUBLE_EPSILON

```
#define DOUBLE_EPSILON 2.220446049250313e-16
```

Definition at line 60 of file [constants_data_type.h](#).

6.4.2.3 DOUBLE_MAX

```
#define DOUBLE_MAX 1.7976931348623158e+308
```

Definition at line 51 of file [constants_data_type.h](#).

6.4.2.4 DOUBLE_MIN

```
#define DOUBLE_MIN -1.7976931348623158e+308
```

Definition at line 50 of file [constants_data_type.h](#).

6.4.2.5 DOUBLE_MIN_POS

```
#define DOUBLE_MIN_POS 2.2250738585072014e-308
```

Definition at line 52 of file [constants_data_type.h](#).

6.4.2.6 DOUBLE_SMALL

```
#define DOUBLE_SMALL 1e-300
```

Definition at line 55 of file [constants_data_type.h](#).

6.4.2.7 FLOAT_BIG

```
#define FLOAT_BIG 1e+30f
```

Definition at line 54 of file [constants_data_type.h](#).

6.4.2.8 FLOAT_EPSILON

```
#define FLOAT_EPSILON 1.1920929e-7f
```

Floating-point machine epsilon constants.

Definition at line 59 of file [constants_data_type.h](#).

6.4.2.9 FLOAT_MAX

```
#define FLOAT_MAX 3.402823466e+38f
```

Definition at line 48 of file [constants_data_type.h](#).

6.4.2.10 FLOAT_MIN

```
#define FLOAT_MIN -3.402823466e+38f
```

Floating-point range and magnitude constants.

Definition at line 47 of file [constants_data_type.h](#).

6.4.2.11 FLOAT_MIN_POS

```
#define FLOAT_MIN_POS 1.175494351e-38f
```

Definition at line 49 of file [constants_data_type.h](#).

6.4.2.12 FLOAT_SMALL

```
#define FLOAT_SMALL 1e-30f
```

Definition at line 53 of file [constants_data_type.h](#).

6.4.2.13 INT_16_MAX

```
#define INT_16_MAX 32767i16
```

Definition at line 15 of file [constants_data_type.h](#).

6.4.2.14 INT_16_MIN

```
#define INT_16_MIN (-32767i16 - 1)
```

Definition at line 14 of file [constants_data_type.h](#).

6.4.2.15 INT_32_MAX

```
#define INT_32_MAX 2147483647i32
```

Definition at line 19 of file [constants_data_type.h](#).

6.4.2.16 INT_32_MIN

```
#define INT_32_MIN (-2147483647i32 - 1)
```

Definition at line 18 of file [constants_data_type.h](#).

6.4.2.17 INT_64_MAX

```
#define INT_64_MAX 9223372036854775807i64
```

Definition at line 23 of file [constants_data_type.h](#).

6.4.2.18 INT_64_MIN

```
#define INT_64_MIN (-9223372036854775807i64 - 1)
```

Definition at line 22 of file [constants_data_type.h](#).

6.4.2.19 INT_8_MAX

```
#define INT_8_MAX 127i8
```

Definition at line 11 of file [constants_data_type.h](#).

6.4.2.20 INT_8_MIN

```
#define INT_8_MIN (-127i8 - 1)
```

Integer type limits.

Definition at line 10 of file [constants_data_type.h](#).

6.4.2.21 LONG_MAX

```
#define LONG_MAX 2147483647L
```

Definition at line 27 of file [constants_data_type.h](#).

6.4.2.22 LONG_MIN

```
#define LONG_MIN (-2147483647L - 1)
```

Definition at line 26 of file [constants_data_type.h](#).

6.4.2.23 UINT_16_MAX

```
#define UINT_16_MAX 0xffffui16
```

Definition at line 17 of file [constants_data_type.h](#).

6.4.2.24 UINT_16_MIN

```
#define UINT_16_MIN 0x0ui16
```

Definition at line 16 of file [constants_data_type.h](#).

6.4.2.25 UINT_32_MAX

```
#define UINT_32_MAX 0xffffffffui32
```

Definition at line 21 of file [constants_data_type.h](#).

6.4.2.26 UINT_32_MIN

```
#define UINT_32_MIN 0x0ui32
```

Definition at line 20 of file [constants_data_type.h](#).

6.4.2.27 UINT_64_MAX

```
#define UINT_64_MAX 0xffffffffffffffffui64
```

Definition at line 25 of file [constants_data_type.h](#).

6.4.2.28 UINT_64_MIN

```
#define UINT_64_MIN 0x0ui64
```

Definition at line 24 of file [constants_data_type.h](#).

6.4.2.29 UINT_8_MAX

```
#define UINT_8_MAX 0xffui8
```

Definition at line 13 of file [constants_data_type.h](#).

6.4.2.30 UINT_8_MIN

```
#define UINT_8_MIN 0x0ui8
```

Definition at line 12 of file [constants_data_type.h](#).

6.4.2.31 ULONG_MAX

```
#define ULONG_MAX 0xffffffffUL
```

Definition at line 29 of file [constants_data_type.h](#).

6.4.2.32 ULONG_MIN

```
#define ULONG_MIN 0x0UL
```

Definition at line 28 of file [constants_data_type.h](#).

6.5 constants_data_type.h

[Go to the documentation of this file.](#)

```

00001
00005
00006 #ifndef CONSTANTS_DATA_TYPE_H
00007 #define CONSTANTS_DATA_TYPE_H
00008
00010 #define INT_8_MIN      (-127i8 - 1)
00011 #define INT_8_MAX      127i8
00012 #define UINT_8_MIN     0x0ui8
00013 #define UINT_8_MAX     0xffui8
00014 #define INT_16_MIN     (-32767i16 - 1)
00015 #define INT_16_MAX     32767i16
00016 #define UINT_16_MIN    0x0ui16
00017 #define UINT_16_MAX    0xffffui16
00018 #define INT_32_MIN     (-2147483647i32 - 1)
00019 #define INT_32_MAX     2147483647i32
00020 #define UINT_32_MIN    0x0ui32
00021 #define UINT_32_MAX    0xffffffffui32
00022 #define INT_64_MIN     (-9223372036854775807i64 - 1)
00023 #define INT_64_MAX     9223372036854775807i64
00024 #define UINT_64_MIN    0x0ui64
00025 #define UINT_64_MAX    0xffffffffffffffffui64
00026 #define LONG_MIN       (-2147483647L - 1)
00027 #define LONG_MAX       2147483647L
00028 #define ULONG_MIN      0x0UL
00029 #define ULONG_MAX      0xffffffffUL
00030
00031 // #define LONG_LONG_MIN  (-9223372036854775807i64 - 1)
00032 // #define LONG_LONG_MAX  9223372036854775807i64
00033 // #define ULONG_LONG_MIN 0x0ui64
00034 // #define ULONG_LONG_MAX 0xffffffffffffffffui64
00035 // #define SCHAR_MIN     -128
00036 // #define SCHAR_MAX     127
00037 // #define UCHAR_MIN     0x00
00038 // #define UCHAR_MAX     0xff
00039 // #define SHORT_MIN     (-32768)
00040 // #define SHORT_MAX     32767
00041 // #define USHORT_MIN    0x00
00042 // #define USHORT_MAX    0xffff
00043 // #define INT_MIN       (-2147483647i - 1)
00044 // #define INT_MAX       2147483647i
00045
00047 #define FLOAT_MIN      -3.402823466e+38f
00048 #define FLOAT_MAX      3.402823466e+38f
00049 #define FLOAT_MIN_POS  1.175494351e-38f
00050 #define DOUBLE_MIN     -1.7976931348623158e+308
00051 #define DOUBLE_MAX     1.7976931348623158e+308
00052 #define DOUBLE_MIN_POS 2.2250738585072014e-308
00053 #define FLOAT_SMALL    1e-30f // a small 32-bit floating-point number
00054 #define FLOAT_BIG      1e+30f // a large 32-bit floating-point number
00055 #define DOUBLE_SMALL   1e-300 // a small 64-bit floating-point number
00056 #define DOUBLE_BIG     1e+300 // a large 64-bit floating-point number
00057
00059 #define FLOAT_EPSILON  1.1920929e-7f
00060 #define DOUBLE_EPSILON 2.220446049250313e-16
00061
00062 #endif // CONSTANTS_DATA_TYPE_H

```

6.6 include/constants_math.h File Reference

Mathematical constants and precomputed root values.

```
graph BT; A[include/constants.h] --> B[include/constants_math.h]
```

- #define MATH_10_PI 31.41592653589793238462643383279502884
- #define MATH_2_PI 6.28318530717958647692528676655900576
- #define MATH_3_PI 9.42477796076937971538793014983850865
- #define MATH_4_PI 12.56637061435917295385057353311801153
- #define MATH_5_PI 15.70796326794896619231321691639751442
- #define MATH_6_PI 18.84955592153875943077586029967701730
- #define MATH_7_PI 21.99114857512855266923850368295652018
- #define MATH_8_PI 25.13274122871834590770114706623602307
- #define MATH_9_PI 28.27433388230813914616379044951552595
- #define MATH_CUBRT_10 2.154434690031883814498314677621237
- #define MATH_CUBRT_11 2.223980090569315226645130678662098
- #define MATH_CUBRT_12 2.289428485106663746506683310144580
- #define MATH_CUBRT_13 2.351334687720757266049531608587130
- #define MATH_CUBRT_14 2.410142264175229698253133392427116
- #define MATH_CUBRT_15 2.466212074330469850025338018895126
- #define MATH_CUBRT_16 2.519842099789745937243878870503976
- #define MATH_CUBRT_17 2.571281590658235138846521294908598
- #define MATH_CUBRT_18 2.620741394208896402062691777246072
- #define MATH_CUBRT_19 2.668401648721944852127307967748493
- #define MATH_CUBRT_2 1.259921049894873190666544360283296

- #define MATH_CUBRT_20 2.714417616594906323967961725429631
- #define MATH_CUBRT_21 2.758924176381120751955222658580169
- #define MATH_CUBRT_22 2.802039330655386795143613198888488
- #define MATH_CUBRT_23 2.843866979851565446324457298032939
- #define MATH_CUBRT_24 2.884499140614816603545023099286481
- #define MATH_CUBRT_25 2.924017738212866035496517724823206
- #define MATH_CUBRT_3 1.442249570307408301772511549643240
- #define MATH_CUBRT_4 1.587401051968199361397182656219229
- #define MATH_CUBRT_5 1.709975946676697056147986586438491
- #define MATH_CUBRT_6 1.817120592832139669425828287785407
- #define MATH_CUBRT_7 1.912931182772388938317931206256616
- #define MATH_CUBRT_8 2.0000000000000000000000000000000000
- #define MATH_CUBRT_9 2.080083823051904090561947668902576
- #define MATH e 2.71828182845904509079559829842764884

- #define `MATH_e_POW_2` 7.38905609893065040694182243896648288
- #define `MATH_e_POW_3` 20.08553692318766792368478490971028805
- #define `MATH_e_POW_4` 54.59815003314423620395245961844921112
- #define `MATH_e_POW_5` 148.41315910257659993476408999413251877
- #define `MATH_e_POW_MINUS_1` 0.36787944117144233402427744294982404
- #define `MATH_e_POW_MINUS_2` 0.13533528323661270231781372785917483
- #define `MATH_e_POW_MINUS_3` 0.04978706836786394446248138478949841
- #define `MATH_e_POW_MINUS_4` 0.01831563888873417866864912184610148
- #define `MATH_e_POW_MINUS_5` 0.00673794699908546700084510661099557

Precalculated values of e with various exponents.

- #define `MATH_PI` 3.14159265358979323846264338327950288
- #define `MATH_PI_by_10` 0.31415926535897932384626433832795028

Precalculated values of pi with various multiples and sub-multiples.

- #define `MATH_PI_by_2` 1.57079632679489661923132169163975144
- #define `MATH_PI_by_3` 1.04719755119659774615421446109316762
- #define `MATH_PI_by_4` 0.78539816339744830961566084581987572
- #define `MATH_PI_by_5` 0.62831853071795864769252867665590057
- #define `MATH_PI_by_6` 0.52359877559829887307710723054658381
- #define `MATH_PI_by_7` 0.44879895051282760549466334046850041
- #define `MATH_PI_by_8` 0.39269908169872415480783042290993786
- #define `MATH_PI_by_9` 0.34906585039886591538473815369772254
- #define `MATH_SQRT_10` 3.162277660168379331998893544432718
- #define `MATH_SQRT_11` 3.316624790355399849114932736670686
- #define `MATH_SQRT_12` 3.464101615137754587054892683011744
- #define `MATH_SQRT_13` 3.605551275463989293119221267470495
- #define `MATH_SQRT_14` 3.741657386773941385583748732316549
- #define `MATH_SQRT_15` 3.872983346207416885179265399782399
- #define `MATH_SQRT_16` 4.00000000000000000000000000000000
- #define `MATH_SQRT_17` 4.123105625617660549821409855974077
- #define `MATH_SQRT_18` 4.242640687119285146405066172629094
- #define `MATH_SQRT_19` 4.358898943540673552236981983859615
- #define `MATH_SQRT_2` 1.414213562373095048801688724209698

Precalculated square root values from 2 to 25.

- #define `MATH_SQRT_20` 4.472135954999579392818347337462552
- #define `MATH_SQRT_21` 4.582575694955840006588047193728008
- #define `MATH_SQRT_22` 4.690415759823429554565630113544466
- #define `MATH_SQRT_23` 4.795831523312719541597438064162693
- #define `MATH_SQRT_24` 4.898979485566356196394568149411782
- #define `MATH_SQRT_25` 5.00000000000000000000000000000000
- #define `MATH_SQRT_3` 1.732050807568877293527446341505872
- #define `MATH_SQRT_4` 2.00000000000000000000000000000000
- #define `MATH_SQRT_5` 2.236067977499789696409173668731276
- #define `MATH_SQRT_6` 2.449489742783178098197284074705891
- #define `MATH_SQRT_7` 2.645751311064590590501615753639260
- #define `MATH_SQRT_8` 2.828427124746190097603377448419396
- #define `MATH_SQRT_9` 3.00000000000000000000000000000000
- #define `MATH_TAU` 6.28318530717958647692528676655900576

Precalculated values tau.

6.6.1 Detailed Description

Mathematical constants and precomputed root values.

Provides commonly used values of pi, e, square roots, and cube roots.

Definition in file [constants_math.h](#).

6.6.2 Macro Definition Documentation

6.6.2.1 MATH_10_PI

```
#define MATH_10_PI 31.41592653589793238462643383279502884
```

Definition at line 33 of file [constants_math.h](#).

6.6.2.2 MATH_2_PI

```
#define MATH_2_PI 6.28318530717958647692528676655900576
```

Definition at line 25 of file [constants_math.h](#).

6.6.2.3 MATH_3_PI

```
#define MATH_3_PI 9.42477796076937971538793014983850865
```

Definition at line 26 of file [constants_math.h](#).

6.6.2.4 MATH_4_PI

```
#define MATH_4_PI 12.56637061435917295385057353311801153
```

Definition at line 27 of file [constants_math.h](#).

6.6.2.5 MATH_5_PI

```
#define MATH_5_PI 15.70796326794896619231321691639751442
```

Definition at line 28 of file [constants_math.h](#).

6.6.2.6 MATH_6_PI

```
#define MATH_6_PI 18.84955592153875943077586029967701730
```

Definition at line 29 of file [constants_math.h](#).

6.6.2.7 MATH_7_PI

```
#define MATH_7_PI 21.99114857512855266923850368295652018
```

Definition at line 30 of file [constants_math.h](#).

6.6.2.8 MATH_8_PI

```
#define MATH_8_PI 25.13274122871834590770114706623602307
```

Definition at line 31 of file [constants_math.h](#).

6.6.2.9 MATH_9_PI

```
#define MATH_9_PI 28.27433388230813914616379044951552595
```

Definition at line 32 of file [constants_math.h](#).

6.6.2.10 MATH_CUBRT_10

```
#define MATH_CUBRT_10 2.154434690031883814498314677621237
```

Definition at line 82 of file [constants_math.h](#).

6.6.2.11 MATH_CUBRT_11

```
#define MATH_CUBRT_11 2.223980090569315226645130678662098
```

Definition at line 83 of file [constants_math.h](#).

6.6.2.12 MATH_CUBRT_12

```
#define MATH_CUBRT_12 2.289428485106663746506683310144580
```

Definition at line 84 of file [constants_math.h](#).

6.6.2.13 MATH_CUBRT_13

```
#define MATH_CUBRT_13 2.351334687720757266049531608587130
```

Definition at line 85 of file [constants_math.h](#).

6.6.2.14 MATH_CUBRT_14

```
#define MATH_CUBRT_14 2.410142264175229698253133392427116
```

Definition at line 86 of file [constants_math.h](#).

6.6.2.15 MATH_CUBRT_15

```
#define MATH_CUBRT_15 2.466212074330469850025338018895126
```

Definition at line 87 of file [constants_math.h](#).

6.6.2.16 MATH_CUBRT_16

```
#define MATH_CUBRT_16 2.519842099789745937243878870503976
```

Definition at line 88 of file [constants_math.h](#).

6.6.2.17 MATH_CUBRT_17

```
#define MATH_CUBRT_17 2.571281590658235138846521294908598
```

Definition at line 89 of file [constants_math.h](#).

6.6.2.18 MATH_CUBRT_18

```
#define MATH_CUBRT_18 2.620741394208896402062691777246072
```

Definition at line 90 of file [constants_math.h](#).

6.6.2.19 MATH_CUBRT_19

```
#define MATH_CUBRT_19 2.668401648721944852127307967748493
```

Definition at line 91 of file [constants_math.h](#).

6.6.2.20 MATH_CUBRT_2

```
#define MATH_CUBRT_2 1.259921049894873190666544360283296
```

Precalculated cube root values from 2 to 25.

Definition at line 74 of file [constants_math.h](#).

6.6.2.21 MATH_CUBRT_20

```
#define MATH_CUBRT_20 2.714417616594906323967961725429631
```

Definition at line 92 of file [constants_math.h](#).

6.6.2.22 MATH_CUBRT_21

```
#define MATH_CUBRT_21 2.758924176381120751955222658580169
```

Definition at line 93 of file [constants_math.h](#).

6.6.2.23 MATH_CUBRT_22

```
#define MATH_CUBRT_22 2.802039330655386795143613198888488
```

Definition at line 94 of file [constants_math.h](#).

6.6.2.24 MATH_CUBRT_23

```
#define MATH_CUBRT_23 2.843866979851565446324457298032939
```

Definition at line 95 of file [constants_math.h](#).

6.6.2.25 MATH_CUBRT_24

```
#define MATH_CUBRT_24 2.884499140614816603545023099286481
```

Definition at line 96 of file [constants_math.h](#).

6.6.2.26 MATH_CUBRT_25

```
#define MATH_CUBRT_25 2.924017738212866035496517724823206
```

Definition at line 97 of file [constants_math.h](#).

6.6.2.27 MATH_CUBRT_3

```
#define MATH_CUBRT_3 1.442249570307408301772511549643240
```

Definition at line 75 of file [constants_math.h](#).

6.6.2.28 MATH_CUBRT_4

```
#define MATH_CUBRT_4 1.587401051968199361397182656219229
```

Definition at line 76 of file [constants_math.h](#).

6.6.2.29 MATH_CUBRT_5

```
#define MATH_CUBRT_5 1.709975946676697056147986586438491
```

Definition at line 77 of file [constants_math.h](#).

6.6.2.30 MATH_CUBRT_6

```
#define MATH_CUBRT_6 1.817120592832139669425828287785407
```

Definition at line 78 of file [constants_math.h](#).

6.6.2.31 MATH_CUBRT_7

```
#define MATH_CUBRT_7 1.912931182772388938317931206256616
```

Definition at line 79 of file [constants_math.h](#).

6.6.2.32 MATH_CUBRT_8

```
#define MATH_CUBRT_8 2.00000000000000000000000000000000
```

Definition at line 80 of file [constants_math.h](#).

6.6.2.33 MATH_CUBRT_9

```
#define MATH_CUBRT_9 2.080083823051904090561947668902576
```

Definition at line 81 of file [constants_math.h](#).

6.6.2.34 MATH_e

```
#define MATH_e 2.71828182845904509079559829842764884
```

Definition at line 41 of file [constants_math.h](#).

6.6.2.35 MATH_e_POW_2

```
#define MATH_e_POW_2 7.38905609893065040694182243896648288
```

Definition at line 42 of file [constants_math.h](#).

6.6.2.36 MATH_e_POW_3

```
#define MATH_e_POW_3 20.08553692318766792368478490971028805
```

Definition at line 43 of file [constants_math.h](#).

6.6.2.37 MATH_e_POW_4

```
#define MATH_e_POW_4 54.59815003314423620395245961844921112
```

Definition at line 44 of file [constants_math.h](#).

6.6.2.38 MATH_e_POW_5

```
#define MATH_e_POW_5 148.41315910257659993476408999413251877
```

Definition at line 45 of file [constants_math.h](#).

6.6.2.39 MATH_e_POW_MINUS_1

```
#define MATH_e_POW_MINUS_1 0.36787944117144233402427744294982404
```

Definition at line 40 of file [constants_math.h](#).

6.6.2.40 MATH_e_POW_MINUS_2

```
#define MATH_e_POW_MINUS_2 0.13533528323661270231781372785917483
```

Definition at line 39 of file [constants_math.h](#).

6.6.2.41 MATH_e_POW_MINUS_3

```
#define MATH_e_POW_MINUS_3 0.04978706836786394446248138478949841
```

Definition at line 38 of file [constants_math.h](#).

6.6.2.42 MATH_e_POW_MINUS_4

```
#define MATH_e_POW_MINUS_4 0.01831563888873417866864912184610148
```

Definition at line 37 of file [constants_math.h](#).

6.6.2.43 MATH_e_POW_MINUS_5

```
#define MATH_e_POW_MINUS_5 0.00673794699908546700084510661099557
```

Precalculated values of e with various exponents.

Definition at line 36 of file [constants_math.h](#).

6.6.2.44 MATH_PI

```
#define MATH_PI 3.14159265358979323846264338327950288
```

Definition at line 24 of file [constants_math.h](#).

6.6.2.45 MATH_PI_by_10

```
#define MATH_PI_by_10 0.31415926535897932384626433832795028
```

Precalculated values of pi with various multiples and sub-multiples.

Definition at line 15 of file [constants_math.h](#).

6.6.2.46 MATH_PI_by_2

```
#define MATH_PI_by_2 1.57079632679489661923132169163975144
```

Definition at line 23 of file [constants_math.h](#).

6.6.2.47 MATH_PI_by_3

```
#define MATH_PI_by_3 1.04719755119659774615421446109316762
```

Definition at line 22 of file [constants_math.h](#).

6.6.2.48 MATH_PI_by_4

```
#define MATH_PI_by_4 0.78539816339744830961566084581987572
```

Definition at line 21 of file [constants_math.h](#).

6.6.2.49 MATH_PI_by_5

```
#define MATH_PI_by_5 0.62831853071795864769252867665590057
```

Definition at line 20 of file [constants_math.h](#).

6.6.2.50 MATH_PI_by_6

```
#define MATH_PI_by_6 0.52359877559829887307710723054658381
```

Definition at line 19 of file [constants_math.h](#).

6.6.2.51 MATH_PI_by_7

```
#define MATH_PI_by_7 0.44879895051282760549466334046850041
```

Definition at line 18 of file [constants_math.h](#).

6.6.2.52 MATH_PI_by_8

```
#define MATH_PI_by_8 0.39269908169872415480783042290993786
```

Definition at line 17 of file [constants_math.h](#).

6.6.2.53 MATH_PI_by_9

```
#define MATH_PI_by_9 0.34906585039886591538473815369772254
```

Definition at line 16 of file [constants_math.h](#).

6.6.2.54 MATH_SQRT_10

```
#define MATH_SQRT_10 3.162277660168379331998893544432718
```

Definition at line 56 of file [constants_math.h](#).

6.6.2.55 MATH_SQRT_11

```
#define MATH_SQRT_11 3.316624790355399849114932736670686
```

Definition at line 57 of file [constants_math.h](#).

6.6.2.56 MATH_SQRT_12

```
#define MATH_SQRT_12 3.464101615137754587054892683011744
```

Definition at line 58 of file [constants_math.h](#).

6.6.2.57 MATH_SQRT_13

```
#define MATH_SQRT_13 3.605551275463989293119221267470495
```

Definition at line 59 of file [constants_math.h](#).

6.6.2.58 MATH_SQRT_14

```
#define MATH_SQRT_14 3.741657386773941385583748732316549
```

Definition at line 60 of file [constants_math.h](#).

6.6.2.59 MATH_SQRT_15

```
#define MATH_SQRT_15 3.872983346207416885179265399782399
```

Definition at line 61 of file [constants_math.h](#).

6.6.2.60 MATH_SQRT_16

```
#define MATH_SQRT_16 4.00000000000000000000000000000000
```

Definition at line 62 of file [constants_math.h](#).

6.6.2.61 MATH_SQRT_17

```
#define MATH_SQRT_17 4.123105625617660549821409855974077
```

Definition at line 63 of file [constants_math.h](#).

6.6.2.62 MATH_SQRT_18

```
#define MATH_SQRT_18 4.242640687119285146405066172629094
```

Definition at line 64 of file [constants_math.h](#).

6.6.2.63 MATH_SQRT_19

```
#define MATH_SQRT_19 4.358898943540673552236981983859615
```

Definition at line 65 of file [constants_math.h](#).

6.6.2.64 MATH_SQRT_2

```
#define MATH_SQRT_2 1.414213562373095048801688724209698
```

Precalculated square root values from 2 to 25.

Definition at line 48 of file [constants_math.h](#).

6.6.2.65 MATH_SQRT_20

```
#define MATH_SQRT_20 4.472135954999579392818347337462552
```

Definition at line 66 of file [constants_math.h](#).

6.6.2.66 MATH_SQRT_21

```
#define MATH_SQRT_21 4.582575694955840006588047193728008
```

Definition at line 67 of file [constants_math.h](#).

6.6.2.67 MATH_SQRT_22

```
#define MATH_SQRT_22 4.690415759823429554565630113544466
```

Definition at line 68 of file [constants_math.h](#).

6.6.2.68 MATH_SQRT_23

```
#define MATH_SQRT_23 4.795831523312719541597438064162693
```

Definition at line 69 of file [constants_math.h](#).

6.6.2.69 MATH_SQRT_24

```
#define MATH_SQRT_24 4.898979485566356196394568149411782
```

Definition at line 70 of file [constants_math.h](#).

6.6.2.70 MATH_SQRT_25

```
#define MATH_SQRT_25 5.00000000000000000000000000000000
```

Definition at line 71 of file [constants_math.h](#).

6.6.2.71 MATH_SQRT_3

```
#define MATH_SQRT_3 1.732050807568877293527446341505872
```

Definition at line 49 of file [constants_math.h](#).

6.6.2.72 MATH_SQRT_4

```
#define MATH_SQRT_4 2.00000000000000000000000000000000
```

Definition at line 50 of file [constants_math.h](#).

6.6.2.73 MATH_SQRT_5

```
#define MATH_SQRT_5 2.236067977499789696409173668731276
```

Definition at line 51 of file [constants_math.h](#).

6.6.2.74 MATH_SQRT_6

```
#define MATH_SQRT_6 2.449489742783178098197284074705891
```

Definition at line 52 of file [constants_math.h](#).

6.6.2.75 MATH_SQRT_7

```
#define MATH_SQRT_7 2.645751311064590590501615753639260
```

Definition at line 53 of file [constants_math.h](#).

6.6.2.76 MATH_SQRT_8

```
#define MATH_SQRT_8 2.828427124746190097603377448419396
```

Definition at line 54 of file [constants_math.h](#).

6.6.2.77 MATH_SQRT_9

```
#define MATH_SQRT_9 3.00000000000000000000000000000000
```

Definition at line 55 of file [constants_math.h](#).

6.6.2.78 MATH_TAU

```
#define MATH_TAU 6.28318530717958647692528676655900576
```

Precalculated values tau.

Definition at line 12 of file [constants_math.h](#).

6.7 constants_math.h

[Go to the documentation of this file.](#)

```
00001
00007
00008 #ifndef CONSTANTS_MATH_H
00009 #define CONSTANTS_MATH_H
00010
00012 #define MATH_TAU          6.28318530717958647692528676655900576
00013
00015 #define MATH_PI_by_10    0.31415926535897932384626433832795028
00016 #define MATH_PI_by_9     0.34906585039886591538473815369772254
00017 #define MATH_PI_by_8     0.39269908169872415480783042290993786
00018 #define MATH_PI_by_7     0.44879895051282760549466334046850041
00019 #define MATH_PI_by_6     0.52359877559829887307710723054658381
00020 #define MATH_PI_by_5     0.62831853071795864769252867665590057
00021 #define MATH_PI_by_4     0.78539816339744830961566084581987572
00022 #define MATH_PI_by_3     1.04719755119659774615421446109316762
00023 #define MATH_PI_by_2     1.57079632679489661923132169163975144
00024 #define MATH_PI          3.14159265358979323846264338327950288
00025 #define MATH_2_PI        6.28318530717958647692528676655900576
00026 #define MATH_3_PI        9.42477796076937971538793014983850865
00027 #define MATH_4_PI        12.56637061435917295385057353311801153
00028 #define MATH_5_PI        15.70796326794896619231321691639751442
00029 #define MATH_6_PI        18.84955592153875943077586029967701730
00030 #define MATH_7_PI        21.99114857512855266923850368295652018
00031 #define MATH_8_PI        25.13274122871834590770114706623602307
00032 #define MATH_9_PI        28.27433388230813914616379044951552595
00033 #define MATH_10_PI       31.41592653589793238462643383279502884
00034
00036 #define MATH_e_POW_MINUS_5 0.00673794699908546700084510661099557
00037 #define MATH_e_POW_MINUS_4 0.01831563888873417866864912184610148
00038 #define MATH_e_POW_MINUS_3 0.04978706836786394446248138478949841
00039 #define MATH_e_POW_MINUS_2 0.13533528323661270231781372785917483
00040 #define MATH_e_POW_MINUS_1 0.36787944117144233402427744294982404
00041 #define MATH_e            2.71828182845904509079559829842746884
00042 #define MATH_e_POW_2      7.38905609893065040694182243896648288
00043 #define MATH_e_POW_3      20.08553692318766792368478490971028805
00044 #define MATH_e_POW_4      54.59815003314423620395245961844921112
00045 #define MATH_e_POW_5      148.41315910257659993476408999413251877
00046
00048 #define MATH_SQRT_2       1.414213562373095048801688724209698
00049 #define MATH_SQRT_3       1.732050807568877293527446341505872
00050 #define MATH_SQRT_4       2.00000000000000000000000000000000
00051 #define MATH_SQRT_5       2.236067977499789696409173668731276
00052 #define MATH_SQRT_6       2.449489742783178098197284074705891
00053 #define MATH_SQRT_7       2.645751311064590590501615753639260
00054 #define MATH_SQRT_8       2.828427124746190097603377448419396
00055 #define MATH_SQRT_9       3.00000000000000000000000000000000
00056 #define MATH_SQRT_10      3.162277660168379331998893544432718
00057 #define MATH_SQRT_11      3.316624790355399849114932736670686
00058 #define MATH_SQRT_12      3.464101615137754587054892683011744
00059 #define MATH_SQRT_13      3.605551275463989293119221267470495
00060 #define MATH_SQRT_14      3.741657386773941385583748732316549
00061 #define MATH_SQRT_15      3.872983346207416885179265399782399
00062 #define MATH_SQRT_16      4.00000000000000000000000000000000
00063 #define MATH_SQRT_17      4.123105625617660549821409855974077
```

```

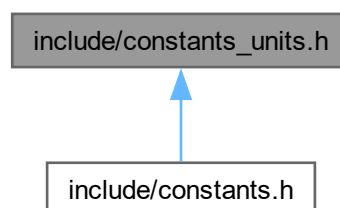
00064 #define MATH_SQRT_18 4.242640687119285146405066172629094
00065 #define MATH_SQRT_19 4.358898943540673552236981983859615
00066 #define MATH_SQRT_20 4.472135954999579392818347337462552
00067 #define MATH_SQRT_21 4.582575694955840006588047193728008
00068 #define MATH_SQRT_22 4.690415759823429554565630113544466
00069 #define MATH_SQRT_23 4.795831523312719541597438064162693
00070 #define MATH_SQRT_24 4.898979485566356196394568149411782
00071 #define MATH_SQRT_25 5.00000000000000000000000000000000
00072
00074 #define MATH_CUBRT_2 1.259921049894873190666544360283296
00075 #define MATH_CUBRT_3 1.442249570307408301772511549643240
00076 #define MATH_CUBRT_4 1.587401051968199361397182656219229
00077 #define MATH_CUBRT_5 1.709975946676697056147986586438491
00078 #define MATH_CUBRT_6 1.817120592832139669425828287785407
00079 #define MATH_CUBRT_7 1.912931182772388938317931206256616
00080 #define MATH_CUBRT_8 2.00000000000000000000000000000000
00081 #define MATH_CUBRT_9 2.080083823051904090561947668902576
00082 #define MATH_CUBRT_10 2.154434690031883814498314677621237
00083 #define MATH_CUBRT_11 2.223980090569315226645130678662098
00084 #define MATH_CUBRT_12 2.289428485106663746506683310144580
00085 #define MATH_CUBRT_13 2.351334687720757266049531608587130
00086 #define MATH_CUBRT_14 2.410142264175229698253133392427116
00087 #define MATH_CUBRT_15 2.466212074330469850025338018895126
00088 #define MATH_CUBRT_16 2.519842099789745937243878870503976
00089 #define MATH_CUBRT_17 2.571281590658235138846521294908598
00090 #define MATH_CUBRT_18 2.620741394208896402062691777246072
00091 #define MATH_CUBRT_19 2.668401648721944852127307967748493
00092 #define MATH_CUBRT_20 2.714417616594906323967961725429631
00093 #define MATH_CUBRT_21 2.758924176381120751955222658580169
00094 #define MATH_CUBRT_22 2.802039330655386795143613198888488
00095 #define MATH_CUBRT_23 2.843866979851565446324457298032939
00096 #define MATH_CUBRT_24 2.884499140614816603545023099286481
00097 #define MATH_CUBRT_25 2.924017738212866035496517724823206
00098
00099 #endif // CONSTANTS_MATH_H

```

6.8 include/constants_units.h File Reference

Unit conversion macros and SI/binary prefixes.

This graph shows which files directly or indirectly include this file:



Macros

- #define [ARCMINUT_2_ARCSECOND\(x\)](#)
- #define [ARCMINUT_2_DEG\(x\)](#)
- #define [ARCSECOND_2_ARCMINUT\(x\)](#)
- #define [ARCSECOND_2_DEG\(x\)](#)
- #define [ATTO](#) 1e-18
- #define [C_2_F\(x\)](#)

Temperature conversion macros.

- #define [C_2_K\(x\)](#)
- #define [C_2_R\(x\)](#)
- #define [CENTI](#) 1e-2
- #define [DAYS_2_HR\(x\)](#)
- #define [DAYS_2_MIN\(x\)](#)
- #define [DAYS_2_SEC\(x\)](#)
- #define [DECI](#) 1e-1
- #define [DEG_2_ARCMINUT\(x\)](#)
- #define [DEG_2_ARCSECOND\(x\)](#)
- #define [DEG_2_RAD\(x\)](#)
- #define [DEGPS_2_RADPS\(x\)](#)
- #define [DEGPS_2_RPM\(x\)](#)
- #define [DEKA](#) 1e1
- #define [EXA](#) 1e18
- #define [EXBI](#) (1 << 60)
- #define [F_2_C\(x\)](#)
- #define [F_2_K\(x\)](#)
- #define [F_2_R\(x\)](#)
- #define [FATHOM_2_M\(x\)](#)
- #define [FATHOM_2_YARD\(x\)](#)
- #define [FEMTO](#) 1e-15
- #define [FT_2_IN\(x\)](#)
- #define [FT_2_M\(x\)](#)
- #define [FT_2_YARD\(x\)](#)
- #define [GIBI](#) (1 << 30)
- #define [GIGA](#) 1e9
- #define [HECTO](#) 1e2
- #define [HR_2_MIN\(x\)](#)
- #define [HR_2_SEC\(x\)](#)
- #define [HZ_2_MS\(x\)](#)
- #define [HZ_2_NS\(x\)](#)
- #define [HZ_2_SEC\(x\)](#)

Frequency and period conversion macros.

- #define [HZ_2_US\(x\)](#)
- #define [IN_2_CM\(x\)](#)
- #define [IN_2_M\(x\)](#)
- #define [K_2_C\(x\)](#)
- #define [K_2_F\(x\)](#)
- #define [K_2_R\(x\)](#)
- #define [KG_2_LB\(x\)](#)
- #define [KG_2_OUNCE\(x\)](#)
- #define [KIBI](#) (1 << 10)

IEC binary prefixes.

- #define [KILO](#) 1e3
- #define [LB_2_KG\(x\)](#)
- #define [LB_2_OUNCE\(x\)](#)
- #define [LEAGUE_2_M\(x\)](#)
- #define [LEAGUE_2_MILE\(x\)](#)
- #define [M_2_FT\(x\)](#)
- #define [M_2_IN\(x\)](#)
- #define [M_2_MILE\(x\)](#)
- #define [MEBI](#) (1 << 20)
- #define [MEGA](#) 1e6

- #define MICRO 1e-6
- #define MIL_2_INCH(x)
- #define MILE_2_FEET(x)
- #define MILE_2_M(x)
- #define MILE_2_YARD(x)
- #define MILLI 1e-3
- #define MIN_2_SEC(x)
- #define MS_2_HZ(x)
- #define MS_2_SEC(x)
- #define NANO 1e-9
- #define NAUT_MILE_2_M(x)
- #define NS_2_HZ(x)
- #define NS_2_SEC(x)
- #define OUNCE_2_KG(x)
- #define OUNCE_2_LB(x)
- #define PEBI (1 << 50)
- #define PETA 1e15
- #define PICO 1e-12
- #define QUECTO 1e-30
- #define QUETTA 1e30

SI decimal prefixes.

- #define R_2_C(x)
- #define R_2_F(x)
- #define R_2_K(x)
- #define RAD_2_DEG(x)

Angle conversion macros.

- #define RADPS_2_DEGPS(x)
- #define RADPS_2_RPM(x)
- #define RONNA 1e27
- #define RONTA 1e-27
- #define RPM_2_DEGPS(x)
- #define RPM_2_RADPS(x)

Rotational speed conversion macros.

- #define SEC_2_HZ(x)
- #define SEC_2_MS(x)
- #define SEC_2_NS(x)

Time conversion macros.

- #define SEC_2_US(x)
- #define TEBI (1 << 40)
- #define TERA 1e12
- #define THOUGH_2_INCH(x)

Length conversion macros.

- #define TN_2_KG(x)

Mass conversion macros.

- #define US_2_HZ(x)
- #define US_2_SEC(x)
- #define YARD_2_FT(x)
- #define YARD_2_M(x)
- #define YOBI (1 << 80)
- #define YOCTO 1e-24
- #define YOTTA 1e24
- #define ZEBI (1 << 70)
- #define ZEPTO 1e-21
- #define ZETTA 1e21

6.8.1 Detailed Description

Unit conversion macros and SI/binary prefixes.

Definition in file [constants_units.h](#).

6.8.2 Macro Definition Documentation

6.8.2.1 ARCMINUT_2_ARCSECOND

```
#define ARCMINUT_2_ARCSECOND(  
    x)
```

Value:

$(x * 60.0)$

Definition at line 83 of file [constants_units.h](#).

6.8.2.2 ARCMINUT_2_DEG

```
#define ARCMINUT_2_DEG(  
    x)
```

Value:

$(x / 60.0)$

Definition at line 84 of file [constants_units.h](#).

6.8.2.3 ARCSECOND_2_ARCMINUT

```
#define ARCSECOND_2_ARCMINUT(  
    x)
```

Value:

$(x / 60.0)$

Definition at line 85 of file [constants_units.h](#).

6.8.2.4 ARCSECOND_2_DEG

```
#define ARCSECOND_2_DEG(  
    x)
```

Value:

$(x / 3600.0)$

Definition at line 86 of file [constants_units.h](#).

6.8.2.5 ATTO

```
#define ATTO 1e-18
```

Definition at line 116 of file [constants_units.h](#).

6.8.2.6 C_2_F

```
#define C_2_F(  
    x)
```

Value:

```
(1.8 * x + 32.0)
```

Temperature conversion macros.

Definition at line 65 of file [constants_units.h](#).

6.8.2.7 C_2_K

```
#define C_2_K(  
    x)
```

Value:

```
(x + 273.15)
```

Definition at line 66 of file [constants_units.h](#).

6.8.2.8 C_2_R

```
#define C_2_R(  
    x)
```

Value:

```
(0.8 * x)
```

Definition at line 67 of file [constants_units.h](#).

6.8.2.9 CENTI

```
#define CENTI 1e-2
```

Definition at line 110 of file [constants_units.h](#).

6.8.2.10 DAYS_2_HR

```
#define DAYS_2_HR(  
    x)
```

Value:

```
(x * 24.0)
```

Definition at line 50 of file [constants_units.h](#).

6.8.2.11 DAYS_2_MIN

```
#define DAYS_2_MIN(  
    x)
```

Value:

(x * 1440.0)

Definition at line 51 of file [constants_units.h](#).

6.8.2.12 DAYS_2_SEC

```
#define DAYS_2_SEC(  
    x)
```

Value:

(x * 86400.0)

Definition at line 52 of file [constants_units.h](#).

6.8.2.13 DECI

```
#define DECI 1e-1
```

Definition at line 109 of file [constants_units.h](#).

6.8.2.14 DEG_2_ARCMINUT

```
#define DEG_2_ARCMINUT(  
    x)
```

Value:

(x * 60.0)

Definition at line 81 of file [constants_units.h](#).

6.8.2.15 DEG_2_ARCSECOND

```
#define DEG_2_ARCSECOND(  
    x)
```

Value:

(x * 3600.0)

Definition at line 82 of file [constants_units.h](#).

6.8.2.16 DEG_2_RAD

```
#define DEG_2_RAD(  
    x)
```

Value:

```
(x * 0.017453292519943295769236907684886127)
```

Definition at line 80 of file [constants_units.h](#).

6.8.2.17 DEGPS_2_RADPS

```
#define DEGPS_2_RADPS(  
    x)
```

Value:

```
(x * 0.017453292519943295769236907684886127)
```

Definition at line 94 of file [constants_units.h](#).

6.8.2.18 DEGPS_2_RPM

```
#define DEGPS_2_RPM(  
    x)
```

Value:

```
(x * 0.166666666666666666666666666666666667)
```

Definition at line 93 of file [constants_units.h](#).

6.8.2.19 DEKA

```
#define DEKA 1e1
```

Definition at line 108 of file [constants_units.h](#).

6.8.2.20 EXA

```
#define EXA 1e18
```

Definition at line 101 of file [constants_units.h](#).

6.8.2.21 EXBI

```
#define EXBI (1 << 60)
```

Definition at line 128 of file [constants_units.h](#).

6.8.2.22 F_2_C

```
#define F_2_C(  
    x)
```

Value:

$((x - 32.0) / 1.8)$

Definition at line 68 of file [constants_units.h](#).

6.8.2.23 F_2_K

```
#define F_2_K(  
    x)
```

Value:

$((x - 32.0) / 1.8 + 273.15)$

Definition at line 69 of file [constants_units.h](#).

6.8.2.24 F_2_R

```
#define F_2_R(  
    x)
```

Value:

$((x - 32.0) / 2.25)$

Definition at line 70 of file [constants_units.h](#).

6.8.2.25 FATHOM_2_M

```
#define FATHOM_2_M(  
    x)
```

Value:

$(x * 1.8288)$

Definition at line 28 of file [constants_units.h](#).

6.8.2.26 FATHOM_2_YARD

```
#define FATHOM_2_YARD(  
    x)
```

Value:

$(x * 2.0)$

Definition at line 27 of file [constants_units.h](#).

6.8.2.27 FEMTO

```
#define FEMTO 1e-15
```

Definition at line 115 of file [constants_units.h](#).

6.8.2.28 FT_2_IN

```
#define FT_2_IN(  
    x)
```

Value:

```
(x * 12.0)
```

Definition at line 15 of file [constants_units.h](#).

6.8.2.29 FT_2_M

```
#define FT_2_M(  
    x)
```

Value:

```
(x * 0.3048)
```

Definition at line 16 of file [constants_units.h](#).

6.8.2.30 FT_2_YARD

```
#define FT_2_YARD(  
    x)
```

Value:

```
(x / 3.0)
```

Definition at line 20 of file [constants_units.h](#).

6.8.2.31 GIBI

```
#define GIBI (1 << 30)
```

Definition at line 125 of file [constants_units.h](#).

6.8.2.32 GIGA

```
#define GIGA 1e9
```

Definition at line 104 of file [constants_units.h](#).

6.8.2.33 HECTO

```
#define HECTO 1e2
```

Definition at line 107 of file [constants_units.h](#).

6.8.2.34 HR_2_MIN

```
#define HR_2_MIN(  
    x)
```

Value:

```
(x * 60.0)
```

Definition at line 49 of file [constants_units.h](#).

6.8.2.35 HR_2_SEC

```
#define HR_2_SEC(  
    x)
```

Value:

```
(x * 3600.0)
```

Definition at line 48 of file [constants_units.h](#).

6.8.2.36 HZ_2_MS

```
#define HZ_2_MS(  
    x)
```

Value:

```
(1000.0 / x)
```

Definition at line 56 of file [constants_units.h](#).

6.8.2.37 HZ_2_NS

```
#define HZ_2_NS(  
    x)
```

Value:

```
(1000000000.0 / x)
```

Definition at line 58 of file [constants_units.h](#).

6.8.2.38 HZ_2_SEC

```
#define HZ_2_SEC(  
    x)
```

Value:

$(1.0 / x)$

Frequency and period conversion macros.

Definition at line 55 of file [constants_units.h](#).

6.8.2.39 HZ_2_US

```
#define HZ_2_US(  
    x)
```

Value:

$(1000000.0 / x)$

Definition at line 57 of file [constants_units.h](#).

6.8.2.40 IN_2_CM

```
#define IN_2_CM(  
    x)
```

Value:

$(x * 2.54)$

Definition at line 12 of file [constants_units.h](#).

6.8.2.41 IN_2_M

```
#define IN_2_M(  
    x)
```

Value:

$(x * 0.0254)$

Definition at line 13 of file [constants_units.h](#).

6.8.2.42 K_2_C

```
#define K_2_C(  
    x)
```

Value:

$(x - 273.15)$

Definition at line 71 of file [constants_units.h](#).

6.8.2.43 K_2_F

```
#define K_2_F(  
    x)
```

Value:

$(1.8 * x - 459.67)$

Definition at line 72 of file [constants_units.h](#).

6.8.2.44 K_2_R

```
#define K_2_R(  
    x)
```

Value:

$(0.8 * x - 218.52)$

Definition at line 73 of file [constants_units.h](#).

6.8.2.45 KG_2_LB

```
#define KG_2_LB(  
    x)
```

Value:

$(x / 0.45359237)$

Definition at line 34 of file [constants_units.h](#).

6.8.2.46 KG_2_OUNCE

```
#define KG_2_OUNCE(  
    x)
```

Value:

$(x * 35.273961949580412915675808215204)$

Definition at line 38 of file [constants_units.h](#).

6.8.2.47 KIBI

```
#define KIBI (1 << 10)
```

IEC binary prefixes.

Definition at line 123 of file [constants_units.h](#).

6.8.2.48 KILO

```
#define KILO 1e3
```

Definition at line 106 of file [constants_units.h](#).

6.8.2.49 LB_2_KG

```
#define LB_2_KG(  
    x)
```

Value:

```
(x * 0.45359237)
```

Definition at line 33 of file [constants_units.h](#).

6.8.2.50 LB_2_OUNCE

```
#define LB_2_OUNCE(  
    x)
```

Value:

```
(x * 0.0625)
```

Definition at line 35 of file [constants_units.h](#).

6.8.2.51 LEAGUE_2_M

```
#define LEAGUE_2_M(  
    x)
```

Value:

```
(x * 4800.0)
```

Definition at line 26 of file [constants_units.h](#).

6.8.2.52 LEAGUE_2_MILE

```
#define LEAGUE_2_MILE(  
    x)
```

Value:

```
(x * 3.0)
```

Definition at line 25 of file [constants_units.h](#).

6.8.2.53 M_2_FT

```
#define M_2_FT(  
    x)
```

Value:

(x * 3.2808398950131233595800524934383202)

Definition at line 17 of file [constants_units.h](#).

6.8.2.54 M_2_IN

```
#define M_2_IN(  
    x)
```

Value:

(x * 39.370078740157480314960629921259843)

Definition at line 14 of file [constants_units.h](#).

6.8.2.55 M_2_MILE

```
#define M_2_MILE(  
    x)
```

Value:

(x / 1609.344)

Definition at line 24 of file [constants_units.h](#).

6.8.2.56 MEBI

```
#define MEBI (1 << 20)
```

Definition at line 124 of file [constants_units.h](#).

6.8.2.57 MEGA

```
#define MEGA 1e6
```

Definition at line 105 of file [constants_units.h](#).

6.8.2.58 MICRO

```
#define MICRO 1e-6
```

Definition at line 112 of file [constants_units.h](#).

6.8.2.59 MIL_2_INCH

```
#define MIL_2_INCH(  
    x)
```

Value:

(x * 0.001)

Definition at line 11 of file [constants_units.h](#).

6.8.2.60 MILE_2_FEET

```
#define MILE_2_FEET(  
    x)
```

Value:

(x * 5280)

Definition at line 21 of file [constants_units.h](#).

6.8.2.61 MILE_2_M

```
#define MILE_2_M(  
    x)
```

Value:

(x * 1609.344)

Definition at line 23 of file [constants_units.h](#).

6.8.2.62 MILE_2_YARD

```
#define MILE_2_YARD(  
    x)
```

Value:

(x * 1760)

Definition at line 22 of file [constants_units.h](#).

6.8.2.63 MILLI

```
#define MILLI 1e-3
```

Definition at line 111 of file [constants_units.h](#).

6.8.2.64 MIN_2_SEC

```
#define MIN_2_SEC(  
    x)
```

Value:

(x * 60.0)

Definition at line 47 of file [constants_units.h](#).

6.8.2.65 MS_2_HZ

```
#define MS_2_HZ(  
    x)
```

Value:

(1000.0 / x)

Definition at line 60 of file [constants_units.h](#).

6.8.2.66 MS_2_SEC

```
#define MS_2_SEC(  
    x)
```

Value:

(x * 0.001)

Definition at line 44 of file [constants_units.h](#).

6.8.2.67 NANO

```
#define NANO 1e-9
```

Definition at line 113 of file [constants_units.h](#).

6.8.2.68 NAUT_MILE_2_M

```
#define NAUT_MILE_2_M(  
    x)
```

Value:

(x * 1852.0)

Definition at line 29 of file [constants_units.h](#).

6.8.2.69 NS_2_HZ

```
#define NS_2_HZ(  
    x)
```

Value:

```
(1000000000.0 / x)
```

Definition at line 62 of file [constants_units.h](#).

6.8.2.70 NS_2_SEC

```
#define NS_2_SEC(  
    x)
```

Value:

```
(x * 0.000000001)
```

Definition at line 46 of file [constants_units.h](#).

6.8.2.71 OUNCE_2_KG

```
#define OUNCE_2_KG(  
    x)
```

Value:

```
(x * 0.028349523125)
```

Definition at line 37 of file [constants_units.h](#).

6.8.2.72 OUNCE_2_LB

```
#define OUNCE_2_LB(  
    x)
```

Value:

```
(x * 16.0)
```

Definition at line 36 of file [constants_units.h](#).

6.8.2.73 PEBI

```
#define PEBI (1 << 50)
```

Definition at line 127 of file [constants_units.h](#).

6.8.2.74 PETA

```
#define PETA 1e15
```

Definition at line 102 of file [constants_units.h](#).

6.8.2.75 PICO

```
#define PICO 1e-12
```

Definition at line 114 of file [constants_units.h](#).

6.8.2.76 QUECTO

```
#define QUECTO 1e-30
```

Definition at line 120 of file [constants_units.h](#).

6.8.2.77 QUETTA

```
#define QUETTA 1e30
```

SI decimal prefixes.

Definition at line 97 of file [constants_units.h](#).

6.8.2.78 R_2_C

```
#define R_2_C(  
    x)
```

Value:

```
(1.25 * x)
```

Definition at line 74 of file [constants_units.h](#).

6.8.2.79 R_2_F

```
#define R_2_F(  
    x)
```

Value:

```
(2.25 * x + 32.0)
```

Definition at line 75 of file [constants_units.h](#).

6.8.2.80 R_2_K

```
#define R_2_K(  
    x)
```

Value:

```
(1.25 * x + 273.15)
```

Definition at line 76 of file [constants_units.h](#).

6.8.2.81 RAD_2_DEG

```
#define RAD_2_DEG(  
    x)
```

Value:

(x * 57.295779513082322864647721871733665)

Angle conversion macros.

Definition at line 79 of file [constants_units.h](#).

6.8.2.82 RADPS_2_DEGPS

```
#define RADPS_2_DEGPS(  
    x)
```

Value:

(x * 57.295779513082322864647721871733665)

Definition at line 92 of file [constants_units.h](#).

6.8.2.83 RADPS_2_RPM

```
#define RADPS_2_RPM(  
    x)
```

Value:

(x * 9.5492965855137201461330258023509)

Definition at line 91 of file [constants_units.h](#).

6.8.2.84 RONNA

```
#define RONNA 1e27
```

Definition at line 98 of file [constants_units.h](#).

6.8.2.85 RONT0

```
#define RONT0 1e-27
```

Definition at line 119 of file [constants_units.h](#).

6.8.2.86 RPM_2_DEGPS

```
#define RPM_2_DEGPS(  
    x)
```

Value:

(x * 6.0)

Definition at line 90 of file [constants_units.h](#).

6.8.2.87 RPM_2_RADPS

```
#define RPM_2_RADPS(  
    x)
```

Value:

```
(x * 0.10471975511965977461542144610932)
```

Rotational speed conversion macros.

Definition at line 89 of file [constants_units.h](#).

6.8.2.88 SEC_2_HZ

```
#define SEC_2_HZ(  
    x)
```

Value:

```
(1.0 / x)
```

Definition at line 59 of file [constants_units.h](#).

6.8.2.89 SEC_2_MS

```
#define SEC_2_MS(  
    x)
```

Value:

```
(x * 1000.0)
```

Definition at line 43 of file [constants_units.h](#).

6.8.2.90 SEC_2_NS

```
#define SEC_2_NS(  
    x)
```

Value:

```
(x * 1000000000.0)
```

Time conversion macros.

Definition at line 41 of file [constants_units.h](#).

6.8.2.91 SEC_2_US

```
#define SEC_2_US(  
    x)
```

Value:

```
(x * 1000000.0)
```

Definition at line 42 of file [constants_units.h](#).

6.8.2.92 TEBI

```
#define TEBI (1 << 40)
```

Definition at line 126 of file [constants_units.h](#).

6.8.2.93 TERA

```
#define TERA 1e12
```

Definition at line 103 of file [constants_units.h](#).

6.8.2.94 THOUGH_2_INCH

```
#define THOUGH_2_INCH(  
    x)
```

Value:

```
(x * 0.001)
```

Length conversion macros.

Definition at line 10 of file [constants_units.h](#).

6.8.2.95 TN_2_KG

```
#define TN_2_KG(  
    x)
```

Value:

```
(x * 1000.0)
```

Mass conversion macros.

Definition at line 32 of file [constants_units.h](#).

6.8.2.96 US_2_HZ

```
#define US_2_HZ(  
    x)
```

Value:

```
(1000000.0 / x)
```

Definition at line 61 of file [constants_units.h](#).

6.8.2.97 US_2_SEC

```
#define US_2_SEC(  
    x)
```

Value:

(x * 0.000001)

Definition at line 45 of file [constants_units.h](#).

6.8.2.98 YARD_2_FT

```
#define YARD_2_FT(  
    x)
```

Value:

(x * 3.0)

Definition at line 18 of file [constants_units.h](#).

6.8.2.99 YARD_2_M

```
#define YARD_2_M(  
    x)
```

Value:

(x * 0.9144)

Definition at line 19 of file [constants_units.h](#).

6.8.2.100 YOBI

```
#define YOBI (1 << 80)
```

Definition at line 130 of file [constants_units.h](#).

6.8.2.101 YOCTO

```
#define YOCTO 1e-24
```

Definition at line 118 of file [constants_units.h](#).

6.8.2.102 YOTTA

```
#define YOTTA 1e24
```

Definition at line 99 of file [constants_units.h](#).

6.8.2.103 ZEBI

```
#define ZEBI (1 << 70)
```

Definition at line 129 of file [constants_units.h](#).

6.8.2.104 ZEPTO

```
#define ZEPTO 1e-21
```

Definition at line 117 of file [constants_units.h](#).

6.8.2.105 ZETTA

```
#define ZETTA 1e21
```

Definition at line 100 of file [constants_units.h](#).

6.9 constants_units.h

[Go to the documentation of this file.](#)

```
00001
00005
00006 #ifndef CONSTANTS_UNITS_H
00007 #define CONSTANTS_UNITS_H
00008
00010 #define THOUGH_2_INCH(x) (x * 0.001)
00011 #define MIL_2_INCH(x) (x * 0.001)
00012 #define IN_2_CM(x) (x * 2.54)
00013 #define IN_2_M(x) (x * 0.0254)
00014 #define M_2_IN(x) (x * 39.370078740157480314960629921259843)
00015 #define FT_2_IN(x) (x * 12.0)
00016 #define FT_2_M(x) (x * 0.3048)
00017 #define M_2_FT(x) (x * 3.2808398950131233595800524934383202)
00018 #define YARD_2_FT(x) (x * 3.0)
00019 #define YARD_2_M(x) (x * 0.9144)
00020 #define FT_2_YARD(x) (x / 3.0)
00021 #define MILE_2_FEET(x) (x * 5280)
00022 #define MILE_2_YARD(x) (x * 1760)
00023 #define MILE_2_M(x) (x * 1609.344)
00024 #define M_2_MILE(x) (x / 1609.344)
00025 #define LEAGUE_2_MILE(x) (x * 3.0)
00026 #define LEAGUE_2_M(x) (x * 4800.0)
00027 #define FATHOM_2_YARD(x) (x * 2.0)
00028 #define FATHOM_2_M(x) (x * 1.8288)
00029 #define NAUT_MILE_2_M(x) (x * 1852.0)
00030
00032 #define TN_2_KG(x) (x * 1000.0)
00033 #define LB_2_KG(x) (x * 0.45359237)
00034 #define KG_2_LB(x) (x / 0.45359237)
00035 #define LB_2_OUNCE(x) (x * 0.0625)
00036 #define OUNCE_2_LB(x) (x * 16.0)
00037 #define OUNCE_2_KG(x) (x * 0.028349523125)
00038 #define KG_2_OUNCE(x) (x * 35.273961949580412915675808215204)
00039
00041 #define SEC_2_NS(x) (x * 1000000000.0)
00042 #define SEC_2_US(x) (x * 1000000.0)
00043 #define SEC_2_MS(x) (x * 1000.0)
00044 #define MS_2_SEC(x) (x * 0.001)
00045 #define US_2_SEC(x) (x * 0.000001)
00046 #define NS_2_SEC(x) (x * 0.000000001)
00047 #define MIN_2_SEC(x) (x * 60.0)
00048 #define HR_2_SEC(x) (x * 3600.0)
00049 #define HR_2_MIN(x) (x * 60.0)
00050 #define DAYS_2_HR(x) (x * 24.0)
00051 #define DAYS_2_MIN(x) (x * 1440.0)
00052 #define DAYS_2_SEC(x) (x * 86400.0)
```

```

00053
00055 #define HZ_2_SEC(x)      (1.0 / x)
00056 #define HZ_2_MS(x)        (1000.0 / x)
00057 #define HZ_2_US(x)        (1000000.0 / x)
00058 #define HZ_2_NS(x)        (1000000000.0 / x)
00059 #define SEC_2_HZ(x)        (1.0 / x)
00060 #define MS_2_HZ(x)         (1000.0 / x)
00061 #define US_2_HZ(x)         (1000000.0 / x)
00062 #define NS_2_HZ(x)         (1000000000.0 / x)
00063
00065 #define C_2_F(x)           (1.8 * x + 32.0)
00066 #define C_2_K(x)           (x + 273.15)
00067 #define C_2_R(x)           (0.8 * x)
00068 #define F_2_C(x)           ((x - 32.0) / 1.8)
00069 #define F_2_K(x)           ((x - 32.0) / 1.8 + 273.15)
00070 #define F_2_R(x)           ((x - 32.0) / 2.25)
00071 #define K_2_C(x)           (x - 273.15)
00072 #define K_2_F(x)           (1.8 * x - 459.67)
00073 #define K_2_R(x)           (0.8 * x - 218.52)
00074 #define R_2_C(x)           (1.25 * x)
00075 #define R_2_F(x)           (2.25 * x + 32.0)
00076 #define R_2_K(x)           (1.25 * x + 273.15)
00077
00079 #define RAD_2_DEG(x)        (x * 57.295779513082322864647721871733665)
00080 #define DEG_2_RAD(x)        (x * 0.017453292519943295769236907684886127)
00081 #define DEG_2_ARCMINUT(x)  (x * 60.0)
00082 #define DEG_2_ARCSECOND(x) (x * 3600.0)
00083 #define ARCMINUT_2_ARCSECOND(x) (x * 60.0)
00084 #define ARCMINUT_2_DEG(x)  (x / 60.0)
00085 #define ARCSECOND_2_ARCMINUT(x) (x / 60.0)
00086 #define ARCSECOND_2_DEG(x) (x / 3600.0)
00087
00089 #define RPM_2_RADPS(x)      (x * 0.10471975511965977461542144610932)
00090 #define RPM_2_DEGPS(x)      (x * 6.0)
00091 #define RADPS_2_RPM(x)      (x * 9.5492965855137201461330258023509)
00092 #define RADPS_2_DEGPS(x)    (x * 57.295779513082322864647721871733665)
00093 #define DEGPS_2_RPM(x)      (x * 0.16666666666666666666666666666667)
00094 #define DEGPS_2_RADPS(x)    (x * 0.017453292519943295769236907684886127)
00095
00097 #define QUETTA 1e30
00098 #define RONNA 1e27
00099 #define YOTTA 1e24
00100 #define ZETTA 1e21
00101 #define EXA 1e18
00102 #define PETA 1e15
00103 #define TERA 1e12
00104 #define GIGA 1e9
00105 #define MEGA 1e6
00106 #define KILO 1e3
00107 #define HECTO 1e2
00108 #define DEKA 1e1
00109 #define DECI 1e-1
00110 #define CENTI 1e-2
00111 #define MILLI 1e-3
00112 #define MICRO 1e-6
00113 #define NANO 1e-9
00114 #define PICO 1e-12
00115 #define FEMTO 1e-15
00116 #define ATTO 1e-18
00117 #define ZEPTO 1e-21
00118 #define YOCTO 1e-24
00119 #define RONTO 1e-27
00120 #define QUECTO 1e-30
00121
00123 #define KIBI (1 << 10)
00124 #define MEBI (1 << 20)
00125 #define GIBI (1 << 30)
00126 #define TEBI (1 << 40)
00127 #define PEBI (1 << 50)
00128 #define EXBI (1 << 60)
00129 #define ZEBI (1 << 70)
00130 #define YOBI (1 << 80)
00131
00132 #endif // CONSTANTS_UNITS_H

```

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